Dynamic Modeling and Control for the Transition of a Multi-Tiltrotor Aerial Vehicle

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***Abstract*—This paper presents the dynamic modeling and control of the transitional flight phase of a Multi-tiltrotor Aerial Vehicle (MTAV) using both rotors and aerodynamic control surfaces as its effectors. To this end, a six-degree-of-freedom model for the dynamics and kinematics of the aircraft is derived, incorporating the effects of elevators, rudders, and ailerons as control efforts rather than solely as disturbances for the typical multirotor modeling. An expanded control allocation matrix is obtained, and, by solving a constrained optimization problem, the required effectors' commands are computed. In the aircraft's modeling, the angular momentum and rotational inertia of the rotors are taken into account in the rotor arrangement, resulting in a lumped disturbance torque that affects the MTAV's dynamics. A proportional-derivative dynamic inversion and control allocation framework is then formulated to stabilize and make the transition between flight phases with lift force being exchanged from lifters to wings. The overall method is evaluated through numerical simulation.**

***Index Terms*—VTOL, multirotor aerial vehicle, multi-tiltrotor aerial vehicle, flight control.**

I. Introduction

Several UAV configurations have been proposed to combine hovering and forward-flight capabilities. One of these is the multi-tiltrotor aerial vehicle (MTAV) which has tilting mechanisms for one or more rotors, providing additional control authority, with many possibilities of allocation of the forces.

UAV research has grown substantially since [1], particularly toward hybrid VTOL configurations. Recent studies have focused on optimal control allocation and control strategies to tackle the highly coupled nonlinear flight dynamics [2], [3]. In particular, nonlinear aerodynamic effects during transitional flight have been investigated, especially for high angles of attack [4]–[6]. This paper presents a unified framework for the dynamic modeling and control of MTAVs, through a hybrid configuration combining tiltrotors and aerodynamic control surfaces as simultaneous force and moment effectors. The resulting control allocation matrix incorporates all effectors, enabling a unified flight control and allocation strategy for the transitional phase of flight. This approach is motivated by the gap between classical multirotor models, valid only for hover, and fixed-wing models, valid only for cruise flight.

The main contributions of this paper are

- A complete six-degrees-of-freedom (DOF) dynamic model for an MTAV during the transitional phase of flight.
- A dynamic-inversion-based stabilization and control allocation framework for the same flight regime.

The remainder of the text is organized as follows. Section II develops the vehicle dynamic model and defines the tracking error's dynamics. Section III defines the control objective and states the proposed stabilization and control allocation framework. Section IV evaluates the overall scheme using numerical simulation. Finally, Section V concludes the paper.

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II. Problem Statement

This section develops the MTAV dynamic model and derives the tracking-error dynamics. Subsection II-A defines the notation used throughout this work. Subsection II-B and II-C derive the force, moment, kinematic, and dynamic models. Subsection II-D formulates the tracking-error dynamics.

A. Notation

In this paper, Euclidean vectors are represented as $\vec{a} \in \mathbb{R}^n$, scalar variables denoted as italic lowercase letters, $a \in \mathbb{R}$, algebraic vectors are represented by boldface lowercase letters, e.g., $\mathbf{a} = (a_1, a_2, \ldots, a_m) \in \mathbb{R}^m$, and matrices as boldface uppercase letters, such as in $\mathbf{A} \in \mathbb{R}^{m\times n}$. Let the identity matrix be $\mathbf{I}_n$, and the zero matrix be $\mathbf{0}_{n\times m}$. Consider $\mathcal{S}_r \triangleq \{\vec{x}_r, \vec{y}_r, \vec{z}_r\}$ as an orthogonal basis of the three-dimensional space for an inertial frame centered at O. Denote $\mathcal{S}_b \triangleq \{\vec{x}_b, \vec{y}_b, \vec{z}_b\}$ as the body reference frame centered at B, in the same position of the center of mass (CM) of the vehicle. The representation or projection of a Euclidean vector $\vec{v}$ in $\mathcal{S}_r$ is defined by $\mathbf{v}_r = (v_1, v_2, v_3) \in \mathbb{R}^3$. The projection of a vector $\vec{r}$ in $\mathcal{S}_b$ and $\mathcal{S}_r$ is defined as $\mathbf{r}_b$ and $\mathbf{r}_r$, respectively, and the relation between them is given by $\mathbf{r}_r = \mathbf{D}^{r/b}\mathbf{r}_b$, through the attitude matrix. Given the orthogonality property, $(\mathbf{D}^{r/b})^{-1} = (\mathbf{D}^{r/b})^T = \mathbf{D}^{b/r}$. Let $r^{\vec{B/O}}$ be the position of $\mathcal{S}_b$ w.r.t. $\mathcal{S}_r$, which can be represented in $\mathcal{S}_b$ by $\mathbf{r}_b^{B/O}$. Finally, the vector product $\vec{w} \triangleq \vec{r} \times \vec{v}$ is represented as, in $\mathcal{S}_b$ projection, $\mathbf{w}_b = [\mathbf{r}_b\times]\mathbf{v}_b$, where

$$[\mathbf{r}_b\times] \triangleq \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 3},$$

with r_1, r_2, and r_3 denoting the components of $\mathbf{r}_b$.

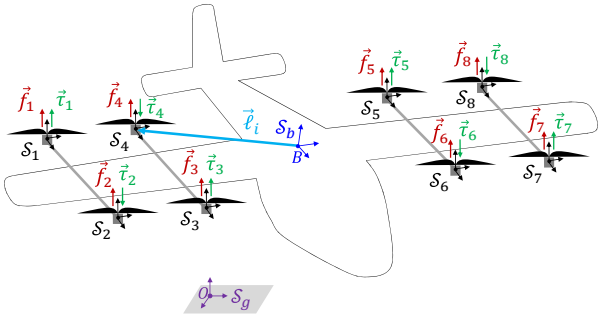


Fig. 1. Schematic representation of the adopted body and inertial reference frames in the MTAV.

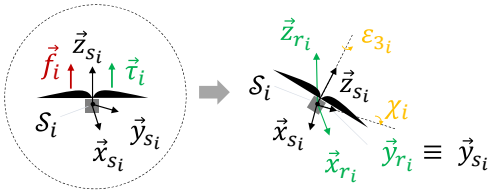


Fig. 2. Schematic representation of the tilting mechanisms and the rotors reference frames.

B. Forces and Moments

Consider the MTAV shown in Fig. 1, which combines hovering and forward-flight capabilities through rotor thrust vectoring and aerodynamic control surfaces.

The MTAV employs eight non-reversible tilting rotors rotating about the $\vec{y}_b$ axis. For $\chi_i = 0$, the rotor thrust is aligned with $\vec{z}_b$. Thus, as shown in Fig. 2, the thrust and reaction torque generated by the ith rotor are given by [2], [3]

$$\vec{f}_i = k_f \varpi_i^2 (\sin\chi_i \vec{x}_b + \cos\chi_i \vec{z}_b), \tag{1}$$

$$\vec{\tau}_i = (-1)^{i+1} k_\tau \varpi_i^2 (\sin\chi_i \vec{x}_b + \cos\chi_i \vec{z}_b), \tag{2}$$

where $\varpi_i \in \mathbb{R}_+$ is the angular speed of the ith rotor, and $k_f \in \mathbb{R}$ and $k_\tau \in \mathbb{R}$ are the propeller's aerodynamic coefficients. Both rotor-speed and tilting-actuator dynamics are modeled as first-order systems, as $\tau_m \dot{\varpi}_i + \varpi_i = k_m \bar{\varpi}_i$, where $\tau_m \in \mathbb{R}$, $k_m \in \mathbb{R}$ and $\bar{\varpi}_i$ is the angular speed command. For the tilting-actuator, $\tau_\chi \dot{\chi}_i + \chi_i = k_\chi \bar{\chi}_i$, where $\tau_\chi \in \mathbb{R}$ and $k_\chi \in \mathbb{R}$ and $\bar{\chi}_i$ is the tilting angular command. Therefore, computing the rotors torque w.r.t. $\mathcal{S}_b$, we obtain,

$$\vec{\tau}^{r/B} = \sum_{i=1}^{8} \vec{\tau}_i + \sum_{i=1}^{8} \vec{l}_i \times \vec{f}_i, \tag{3}$$

and using the numbering and sense of motion defined in Figs. 1 and 2 yields

$$\mathbf{f}_b^r = \begin{bmatrix} \mathbf{\Gamma}_{f_1} & \mathbf{\Gamma}_{f_2} & \dots & \mathbf{\Gamma}_{f_8} \end{bmatrix} \begin{bmatrix} k_f \varpi_1^2 \sin\chi_1 \\ k_f \varpi_1^2 \cos\chi_1 \\ k_f \varpi_2^2 \sin\chi_2 \\ k_f \varpi_2^2 \cos\chi_2 \\ \dots \\ k_f \varpi_8^2 \sin\chi_8 \\ k_f \varpi_8^2 \cos\chi_8 \end{bmatrix}, \tag{4}$$

where $\mathbf{\Gamma}_{f_i}$ can be computed as

$$\mathbf{\Gamma}_{f_i} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \forall i = 1, 2, \dots, 8 \tag{5}$$

and

$$\boldsymbol{\tau}_b^{r/B} = \begin{bmatrix} \mathbf{\Gamma}_{\tau_1} & \mathbf{\Gamma}_{\tau_2} & \dots & \mathbf{\Gamma}_{\tau_8} \end{bmatrix} \begin{bmatrix} k_f \varpi_1^2 \sin\chi_1 \\ k_f \varpi_1^2 \cos\chi_1 \\ k_f \varpi_2^2 \sin\chi_2 \\ k_f \varpi_2^2 \cos\chi_2 \\ \dots \\ k_f \varpi_8^2 \sin\chi_8 \\ k_f \varpi_8^2 \cos\chi_8 \end{bmatrix}, \tag{6}$$

where $\mathbf{\Gamma}_{\tau_i}$ can be computed as,

$$\mathbf{\Gamma}_{\tau_i} = \begin{bmatrix} (-1)^{i+1} k & l_{iy} \\ 0 & -l_{ix} \\ -l_{iy} & (-1)^{i+1} k \end{bmatrix}, \forall i = 1, 2, \dots, 8, \tag{7}$$

where $\mathbf{l}_{ib} = (l_{ix}, l_{iy}, 0)$ is the ith rotor's position represented in the $\mathcal{S}_b$ frame - assumed to be aligned with the height of the center of mass and $k = k_\tau / k_f \in \mathbb{R}$.

Regarding the aerodynamic forces and moments, both the actuator and aircraft dynamics must be modeled. For the former, a first-order model is adopted, as is common in aeronautics literature. For the latter, the development from [1] will be followed closely.

Accordingly, for the three surface actuators, the dynamics are given by $\tau_\delta \dot{\delta}_i + \delta_i = k_\delta \bar{\delta}_i$ where $\tau_\delta \in \mathbb{R}$ and $k_\delta \in \mathbb{R}$ and $\bar{\delta}_i$ is the actuator angular command.

For the aerodynamics, as adapted from [1], let $\mathcal{S}_s \triangleq \{\vec{x}_s, \vec{y}_s, \vec{z}_s\}$ be the stability reference frame also centered at B, in the same position of the center of mass of the vehicle. As seen in Fig. 3, $\mathcal{S}_s$ is obtained from $\mathcal{S}_b$ through a 180-degree turn in the x axis and an α-degree turn in the y axis, where α is the angle of attack of the aircraft. Therefore, $\mathbf{D}^{s/b}$, the attitude matrix of $\mathcal{S}_s$ w.r.t. $\mathcal{S}_b$, can be computed as

$$\mathbf{D}^{s/b} = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & -1 & 0 \\ -\sin\alpha & 0 & -\cos\alpha \end{bmatrix}. \tag{8}$$

As shown in Fig. 3, the airspeed vector $\vec{V}^{b/a}$ forms a sideslip angle β with the $\vec{x}_s$ axis. Hence, its representation in the body frame is $\mathbf{V}_b^{b/a} = \begin{bmatrix} V_a \cos\beta\cos\alpha, V_a \sin\beta, V_a \cos\beta \sin\alpha \end{bmatrix}^T$, where $V_a \triangleq |\vec{V}^{b/a}|$ is airspeed magnitude. The corresponding dynamic pressure is $q_a = \frac{1}{2}\rho {V_a}^2$, where ρ is the air density. Using the nondimensional aerodynamic coefficients, the aerodynamic force $\vec{f}^a$ represented in $\mathcal{S}_s$ is $\mathbf{f}_s^a = \frac{1}{2}\rho {V_a}^2 S [-C_D, C_Y, -C_L]^T$, where S is the MTAV wing planform area.

Let $\boldsymbol{\omega}_b^{b/r} \triangleq (p, q, r) \in \mathbb{R}^3$ denote the angular velocity from the body w.r.t. the inertial frame, expressed in $\mathcal{S}_b$. The aerodynamic coefficients depend on the angular rates and on α and β angles. Following [4]–[6], a function will be used for $C_D(\alpha)$ and $C_L(\alpha)$ that is more adequate for high angles of

attack, instead of the conventional linear approximation, as the MTAV may operate at high angles of attack, especially during the transitional phase. They are given by

$$C_L(\alpha) = (1-\sigma(\alpha))\left(C_{L_0} + C_{L_\alpha}\alpha\right) + \sigma(\alpha)\sin 2\alpha, \quad (9)$$

$$C_D(\alpha) = (1-\sigma(\alpha))\left(C_{D_0} + \frac{(C_{L_0}+C_{L_\alpha}\alpha)^2}{\pi e AR}\right) + \sigma(\alpha)\left(k_{D_1}\sin^2\alpha + k_{D_0}\right), \quad (10)$$

where $k_{D_1} \in \mathbb{R}$ and $k_{D_0} \in \mathbb{R}$ are constants and $\sigma(\alpha)$, also from [4], is given by

$$\sigma(\alpha) = \frac{1 + e^{M(\alpha+\alpha_0)} + e^{-M(\alpha-\alpha_0)}}{\left(1+e^{M(\alpha+\alpha_0)}\right)\left(1+e^{-M(\alpha-\alpha_0)}\right)}, \quad (11)$$

where $M \in \mathbb{R}$ and $\alpha_0 \in \mathbb{R}$ are blending parameters for transition rate and the stall angle, respectively.

In this sense, $\mathbf{f}_s^a$ can be decomposed into controllable and uncontrollable parts, regarding the aerodynamic surfaces,

$$\mathbf{f}_s^{a_{uc}} = q_a S \begin{bmatrix} -C_D(\alpha) - C_{D_q}\frac{c}{2V_a}q \\ C_{Y_0} + C_{Y_\beta}\beta + \frac{c}{2V_a}(C_{Y_p}p + C_{Y_r}r) \\ -C_L(\alpha) - C_{L_q}\frac{c}{2V_a}q \end{bmatrix}, \quad (12)$$

$$\mathbf{f}_s^{a_c} = \mathbf{\Gamma}_{f_\delta}\begin{bmatrix} q_a S\delta_r \\ q_a S\delta_e \\ q_a S\delta_a \end{bmatrix}, \quad (13)$$

where

$$\mathbf{\Gamma}_{f_\delta} = \begin{bmatrix} 0 & -C_{D_{\delta_e}} & 0 \\ C_{Y_{\delta_r}} & 0 & C_{Y_{\delta_a}} \\ 0 & -C_{L_{\delta_e}} & 0 \end{bmatrix}, \quad (14)$$

and c is the mean aerodynamic chord and the derivative aerodynamic coefficients are C_D, C_{D_0}, C_{D_q} and $C_{D_{\delta_e}}$ for the drag force, C_{Y_0}, C_{Y_β}, C_{Y_p}, C_{Y_r}, $C_{Y_{\delta_r}}$ and $C_{Y_{\delta_a}}$ for the lateral force and C_L, C_{L_0}, C_{L_α}, C_{L_q} and $C_{L_{\delta_e}}$ for the lift force. Using the attitude matrix defined in (8), $\mathbf{f}_s^{a_c}$ and $\mathbf{f}_s^{a_{uc}}$ can be represented in the body reference frame as,

$$\mathbf{f}_b^{a_c} = \left(\mathbf{D}^{s/b}\right)^T \mathbf{f}_s^{a_c}, \quad (15)$$

$$\mathbf{f}_b^{a_{uc}} = \left(\mathbf{D}^{s/b}\right)^T \mathbf{f}_s^{a_{uc}}. \quad (16)$$

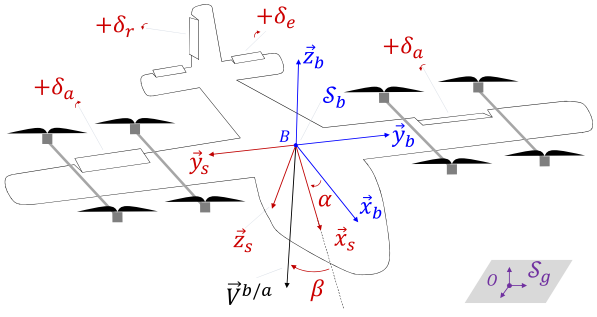


Fig. 3. Schematic representation of the adopted stability reference frame and the aircraft's airspeed.

Adapting [1], a similar derivation for the moments yields

$$\boldsymbol{\tau}_b^{a_{uc}} = q_a S c \begin{bmatrix} C_l(\beta) + \frac{c}{2V_a}\left(C_{l_p}p + C_{l_r}r\right) \\ C_m(\alpha) + C_{m_q}\frac{c}{2V_a}q \\ C_n(\beta) + \frac{c}{2V_a}\left(C_{n_p}p + C_{n_r}r\right) \end{bmatrix}, \quad (17)$$

$$\boldsymbol{\tau}_b^{a_c/B} = \mathbf{\Gamma}_{\tau_\delta}\begin{bmatrix} q_a S\delta_r \\ q_a S\delta_e \\ q_a S\delta_a \end{bmatrix}, \quad (18)$$

where

$$\mathbf{\Gamma}_{\tau_\delta} = \begin{bmatrix} cC_{l_{\delta_r}} & 0 & cC_{l_{\delta_a}} \\ 0 & cC_{m_{\delta_e}} & 0 \\ cC_{n_{\delta_r}} & 0 & cC_{n_{\delta_a}} \end{bmatrix}, \quad (19)$$

and the derivative aerodynamic coefficients are C_l, C_{l_p}, C_{l_r}, $C_{l_{\delta_r}}$ and $C_{l_{\delta_a}}$ for the roll moment, C_m, C_{m_q}, C_{m_0}, $C_{m_{\delta_e}}$ and C_{m_α} for the pitch moment and C_n, C_{n_p}, C_{n_r}, $C_{n_{\delta_r}}$ and $C_{n_{\delta_a}}$ for the yaw moment. A linear approximation is assumed for the stability derivatives dependent on the sideslip angle as $C_l(\beta) = C_{l_0} + C_{l_\beta}\beta$ and $C_n(\beta) = C_{n_0} + C_{n_\beta}\beta$, and also for $C_m(\alpha) = C_{m_0} + C_{m_\alpha}\alpha$.

Following [3] with the notation defined in II-A and using (4), (6), (15) and (18), the total control force and moments are obtained, which, when transformed into matrix form, yields

$$\begin{bmatrix} \mathbf{f}_b^c \\ \boldsymbol{\tau}_b^{c/B} \end{bmatrix} = \mathbf{\Theta}\mathbf{f}, \quad (20)$$

where

$$\mathbf{f} \triangleq \begin{bmatrix} k_f\varpi_1^2\sin\chi_1 \\ k_f\varpi_1^2\cos\chi_1 \\ k_f\varpi_2^2\sin\chi_2 \\ k_f\varpi_2^2\cos\chi_2 \\ \cdots \\ k_f\varpi_8^2\sin\chi_8 \\ k_f\varpi_8^2\cos\chi_8 \\ q_a S\delta_r \\ q_a S\delta_e \\ q_a S\delta_a \end{bmatrix} \in \mathbb{R}^{19\times 1}, \quad (21)$$

and

$$\mathbf{\Theta} \triangleq \begin{bmatrix} \mathbf{\Theta}_f \\ \mathbf{\Theta}_\tau \end{bmatrix} \in \mathbb{R}^{6\times 19} \quad (22)$$

is the total control allocation matrix and is formed by the force and torque control allocation matrices, given by

$$\mathbf{\Theta}_f \triangleq \begin{bmatrix} \mathbf{\Gamma}_{f_1} & \mathbf{\Gamma}_{f_2} & \cdots & \mathbf{\Gamma}_{f_8} & \mathbf{\Gamma}_{f_\delta} \end{bmatrix}, \quad (23)$$

$$\mathbf{\Theta}_\tau \triangleq \begin{bmatrix} \mathbf{\Gamma}_{\tau_1} & \mathbf{\Gamma}_{\tau_2} & \cdots & \mathbf{\Gamma}_{\tau_8} & \mathbf{\Gamma}_{\tau_\delta} \end{bmatrix}, \quad (24)$$

computed with (5), (7), (14) and (19).

C. Rotational and Translational Kinematics and Dynamics

Using the reference frames defined in II-A, here we present the equations of motion for the MTAV. We assume that the vehicle is a rigid-body with constant mass throughout the flight. In addition, the usual flat-earth approach to gravitational forces is used, which leads to $\vec{f}^g = -g\mathbf{e}_3$, where $\mathbf{e}_3 \triangleq (0,0,1)$ is the representation of the $\vec{z}_g$ in $\mathcal{S}_g$ and g is the constant gravitational acceleration.

Following [1], [2] and [3], the attitude kinematics of the body reference frame $\mathcal{S}_b$ w.r.t. the inertial reference frame $\mathcal{S}_r$ is given by

$$\dot{\mathbf{D}}^{b/r} = -\left[\boldsymbol{\omega}_b^{b/r}\times\right]\mathbf{D}^{b/r}, \tag{25}$$

where $\mathbf{D}^{b/r} \in \mathrm{SO}(3)$ is the attitude from the body w.r.t. the inertial frame.

In this paper, due to the tilting mechanics involved in the rotors dynamics, the attitude kinematics of each rotor's own reference frame must also be taken into account. Therefore, let $\mathcal{S}_{s_i}$ be an orthogonal basis of the three-dimensional space for a reference frame centered at B_i, the center of mass of the ith rotor. Let us assume that the tilting movement does not change the position of B_i by means of the actuator position. The axis $\vec{z_{s_i}}$ of the rotor reference frame is aligned with the rotor axis and moves with the tilt angle χ_i. Therefore, the attitude kinematics for each rotor reference frame is given by,

$$\dot{\mathbf{D}}^{s_i/b} = -\left[\boldsymbol{\omega}_{s_i}^{s_i/b}\times\right]\mathbf{D}^{s_i/b}, \tag{26}$$

where $\boldsymbol{\omega}_{s_i}^{s_i/b} \in \mathbb{R}^3$ and $\mathbf{D}^{s_i/b} \in \mathrm{SO}(3)$ are the angular velocity and attitude from the ith rotor w.r.t. the body frame. It is worth mentioning that $\mathcal{S}_{s_i}$ is obtained from $\mathcal{S}_b$ by two rotations, with χ_i along the y axis and ξ_i along the z axis, as shown in Fig. 2, which is given by the rotors rotational movement and its dynamics can be obtained as $\dot{\xi}_i = (-1)^i\bar{\varpi}_i$.

For the angular momentum dynamics from $\mathcal{S}_b$ to $\mathcal{S}_r$ w.r.t. the center of mass B, we obtain

$$\dot{\mathbf{h}}_b^{m/B} + \left[\boldsymbol{\omega}_b^{b/r}\times\right]\mathbf{h}_b^{m/B} = \boldsymbol{\tau}_b^{m/B}, \tag{27}$$

where $\mathbf{h}_b^{m/B} \in \mathbb{R}^3$ is the total angular momentum represented in the body reference frame, $\boldsymbol{\tau}_b^{m/B} \in \mathbb{R}^3$ is the total torque w.r.t. the center of mass B, given by $\boldsymbol{\tau}_b^{m/B} = \boldsymbol{\tau}_b^c + \boldsymbol{\tau}_b^d$, where, in our case, $\boldsymbol{\tau}_b^d \triangleq \boldsymbol{\tau}_b^{a_{uc}}$ is the disturbance torque in the same representation generated by the aerodynamics of the multirotor. Assuming a rigid structure, as shown in [3], $\mathbf{h}_b^{m/B}$ can be computed as

$$\mathbf{h}_b^{m/B} = \mathbf{J}_b^t\boldsymbol{\omega}_b^{b/r} + \sum_{i=1}^{8}\left(\mathbf{D}^{s_i/b}\right)^T\mathbf{J}_{s_i}^i\boldsymbol{\omega}_{s_i}^{s_i/b}, \tag{28}$$

where $\mathbf{J}_b^t$ is the rotational inertia of the multicopter and rotors assembly, that is $\mathbf{J}_b^t \triangleq \mathbf{J}_b + \mathbf{J}_b^r$, which is given by

$$\mathbf{J}_b^r \triangleq \sum_{i=1}^{8}\left(\left(\mathbf{D}^{s_i/b}\right)^T\mathbf{J}_{s_i}^i\mathbf{D}^{s_i/b} - m_i\left[\mathbf{r}_b^{S_i/B}\times\right]^2\right), \tag{29}$$

where m_i is the mass and $\mathbf{J}_{s_i}^i$ is the rotational inertia of each individual rotor. In summary, as it was done in [3], replacing (28) in (27), we obtain the attitude dynamics as

$$\dot{\boldsymbol{\omega}}_b^{b/r} = \left(\mathbf{J}_b^t\right)^{-1}\left[\left(\mathbf{J}_b^t\boldsymbol{\omega}_b^{b/r}\right)\times\right]\boldsymbol{\omega}_b^{b/r} + \left(\mathbf{J}_b^t\right)^{-1}\boldsymbol{\tau}_b^*, \tag{30}$$

where $\boldsymbol{\tau}_b^* = \left(\boldsymbol{\tau}_b^{c/B} + \check{\boldsymbol{\tau}}_b^d\right)$ and $\check{\boldsymbol{\tau}}_b^d \in \mathbb{R}^3$ is a lumped disturbance torque given by

$$\check{\boldsymbol{\tau}}_b^d \triangleq \boldsymbol{\tau}_b^d - \dot{\mathbf{J}}_b^r\boldsymbol{\omega}_b^{b/r} - \sum_{i=1}^{8}\dot{\mathbf{h}}_b^i + \left[\left(\sum_{i=1}^{8}\mathbf{h}_b^i\right)\times\right]\boldsymbol{\omega}_b^{b/r}, \tag{31}$$

where $\mathbf{h}_b^i = \left(\mathbf{D}^{s_i/b}\right)^T\mathbf{J}_{s_i}^i\boldsymbol{\omega}_{s_i}^{s_i/b}$ is the angular momentum of the ith rotor, and the disturbance in our modeling is given by $\boldsymbol{\tau}_b^d = \boldsymbol{\tau}_b^{a_{uc}}$, computed by (17).

For the computation of the attitude matrix of $\mathbf{D}^{s_i/b}$, it is imperative to understand the vectoring and installation of the rotors. First of all, let $\mathcal{S}_{r_i}$ be an orthogonal basis of the three-dimensional space for an inertial frame centered at B_i, as shown in Fig. 2. In addition, $\mathcal{S}_{r_i}$ does not move with the tilting movement and it is assumed that no rotation angle exists around the $\vec{x}_b$ axis. Therefore, let $\mathbf{D}^{r_i/b} \in \mathrm{SO}(3)$ be the attitude from the inertial frame centered in the ith rotor w.r.t. the body frame, which is given by $\mathbf{D}^{r_i/b} = \mathbf{I}_3$, where $\mathbf{I}_3 \in \mathbb{R}^{3\times3}$ is the identity matrix. Then, from the $\mathcal{S}_{r_i}$ reference frame, the $\mathcal{S}_{s_i}$ reference frame can be obtained by the rotation of ε_{2i} and ε_{3i} around the $\vec{y}_b$ and $\vec{z}_b$ axis, respectively, where ε_{3i} exists from the inherent rotational movement of the rotor and ε_{2i} is from the tilting movement of the mechanism. They can be computed as $\varepsilon_{2i} = \chi_i$ and $\dot{\varepsilon}_{3i} = (-1)^i\varpi_i$. Therefore, the attitude matrix $\mathbf{D}^{s_i/b}$ is obtained by

$$\mathbf{D}^{s_i/b} = \mathbf{D}_3(\varepsilon_{3i})\mathbf{D}_2(\varepsilon_{2i})\mathbf{D}^{r_i/b}, \tag{32}$$

where $\mathbf{D}_2(\varepsilon_{2i})$ and $\mathbf{D}_3(\varepsilon_{3i})$ are elementary rotation matrices for euler angles. One can note that, while $\mathbf{D}^{r_i/b}$ is only dependent on the installation of the rotors - in our case, the identity matrix - the product $\mathbf{D}_3(\varepsilon_{3i})\mathbf{D}_2(\varepsilon_{2i})$ is time-dependent. Therefore, $\mathbf{D}^{s_i/b}$ varies with rotors movement along time, following (26).

For the translational movement, as done in [1], a straightforward equation can be obtained by applying Newton's second law, using (16), with vectors represented in $\mathcal{S}_r$, given by

$$\ddot{\mathbf{r}}_r^{B/O} = \frac{1}{m}\left(\mathbf{D}^{b/r}\right)^T\mathbf{f}_b^c - g\mathbf{e}_3 + \frac{1}{m}\mathbf{f}_r^d, \tag{33}$$

where, $\mathbf{v}_r^{B/O} = \dot{\mathbf{r}}_r^{B/O}$ in our modeling, the uncontrollable external force is mainly aerodynamic given by $\mathbf{f}_r^d = \left(\mathbf{D}^{b/r}\right)^T\mathbf{f}_b^d = \left(\mathbf{D}^{b/r}\right)^T\mathbf{f}_b^{a_{uc}}$.

D. Tracking Error Dynamics

Let $\mathcal{S}_{\bar{b}} \triangleq \{\vec{x}_{\bar{b}}, \vec{y}_{\bar{b}}, \vec{z}_{\bar{b}}\}$ be an orthogonal basis of the three-dimensional space for a commanded orientation frame centered at B. Therefore, $\mathbf{D}^{\bar{b}/r} \in \mathrm{SO}(3)$ is the commanded attitude and $\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}$ is the commanded angular velocity for the MTAV.

In the same manner, the attitude and angular-velocity tracking errors can be defined, respectively, as $\tilde{\mathbf{D}} \triangleq \mathbf{D}^{b/\bar{b}} \in \mathrm{SO}(3)$ and $\tilde{\boldsymbol{\omega}} \triangleq \boldsymbol{\omega}_b^{b/\bar{b}} \in \mathbb{R}^3$, which can be rewritten as $\tilde{\mathbf{D}} = \mathbf{D}^{b/r}\left(\mathbf{D}^{\bar{b}/r}\right)^{\mathrm{T}}$ and $\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}_b^{b/r} - \tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}$. Using the same development from II-C, we can obtain $\dot{\tilde{\mathbf{D}}} = -\left[\tilde{\boldsymbol{\omega}}\times\right]\tilde{\mathbf{D}}$, as the tracking error kinematic equation.

Likewise, for the translational movement, let $\bar{\mathbf{r}}_r^{B/O} \in \mathbb{R}^3$ be the commanded position. Therefore, we define the position tracking error as $\tilde{\mathbf{r}} \triangleq \mathbf{r}_r^{B/O} - \bar{\mathbf{r}}_r^{B/O}$, which can be substituted in (33) as

$$\ddot{\tilde{\mathbf{r}}} = \frac{1}{m}\left(\mathbf{D}^{b/r}\right)^T\mathbf{f}_b^c - g\mathbf{e}_3 + \frac{1}{m}\mathbf{f}_r^d - \ddot{\bar{\mathbf{r}}}_r^{B/O}, \tag{34}$$

where, since the actuator's dynamics are fast enough compared to the control loop, one can assume, for (34), $\mathbf{f}_b^c \equiv \bar{\mathbf{f}}_b^c$. In addition, we know that $\mathbf{D}^{b/r} = \tilde{\mathbf{D}}\mathbf{D}^{\bar{b}/r}$, which leads to the dynamic tracking error equations for translational movement, given by

$$\dot{\tilde{\mathbf{r}}} = \tilde{\mathbf{v}}, \tag{35}$$

$$\dot{\tilde{\mathbf{v}}} = \frac{1}{m}\left(\mathbf{D}^{\bar{b}/r}\right)^T \tilde{\mathbf{D}}^T\left(\bar{\mathbf{f}}_b^c + \mathbf{f}_b^d\right) - g\mathbf{e}_3 - \ddot{\bar{\mathbf{r}}}_r^{B/O}, \tag{36}$$

where the second derivative of the commanded position appears, that is why we have to assume that the reference is a smooth function of second order w.r.t. time.

Even though the attitude representation is good for equations' development, when it comes to the control laws' design, a three-dimensional parameterization of $\tilde{\mathbf{D}}$ is more suitable. Therefore, the Euler Angles representation is adopted (see [7], p. 52-65), by assuming the 1-2-3 sequence, we can define $\tilde{\boldsymbol{\alpha}} \triangleq (\tilde{\phi}, \tilde{\theta}, \tilde{\psi})$, where $\tilde{\phi} \in \mathbb{R}$, $\tilde{\theta} \in \mathbb{R}$ and $\tilde{\psi} \in \mathbb{R}$ are the attitude error representation in Euler angles, respectively, corresponding to $\tilde{\mathbf{D}}$, which are related by

$$\tilde{\phi} = -\arctan\left(\tilde{D}_{32}, \tilde{D}_{33}\right), \tag{37}$$

$$\tilde{\theta} = \arcsin\left(\tilde{D}_{31}\right), \tag{38}$$

$$\tilde{\psi} = -\arctan\left(\tilde{D}_{21}, \tilde{D}_{11}\right), \tag{39}$$

with the position of the ith row and jth column of $\tilde{\mathbf{D}}$ being defined by $\tilde{D}_{ij}$. With this representation, the attitude error kinematics can be computed as (see [7], p. 73-76)

$$\dot{\tilde{\boldsymbol{\alpha}}} = \mathbf{A}\tilde{\boldsymbol{\omega}}, \tag{40}$$

where the inverse of the matrix $\mathbf{A}$ is given by

$$\mathbf{A}(\boldsymbol{\alpha})^{-1} = \begin{bmatrix} \cos\psi\cos\theta & \sin\psi & 0 \\ -\sin\psi\cos\theta & \cos\psi & 0 \\ \sin\theta & 0 & 1 \end{bmatrix}. \tag{41}$$

Then, using the same assumption of actuators' dynamics, one can assume that $\boldsymbol{\tau}_b^{c/B} \equiv \bar{\boldsymbol{\tau}}_b^{c/B}$. In addition, by deriving $\tilde{\boldsymbol{\omega}}$ and with the necessary substitutions, the dynamics for the angular velocity tracking error is obtained as

$$\mathbf{J}_b^t\dot{\tilde{\boldsymbol{\omega}}} = \left[\left(\mathbf{J}_b^t\left(\tilde{\boldsymbol{\omega}} + \tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}\right)\right)\times\right]\left(\tilde{\boldsymbol{\omega}} + \tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}\right) + \left[\left(\sum_{i=1}^{8}\mathbf{h}_b^i\right)\times\right]\left(\tilde{\boldsymbol{\omega}} + \tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}\right) - \dot{\mathbf{J}}_b^r\left(\tilde{\boldsymbol{\omega}} + \tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}\right) - \sum_{i=1}^{8}\dot{\mathbf{h}}_b^i + \bar{\boldsymbol{\tau}}_b^{c/B} + \boldsymbol{\tau}_b^d + \mathbf{J}_b^t\left[\tilde{\boldsymbol{\omega}}\times\right]\tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r} - \mathbf{J}_b^t\tilde{\mathbf{D}}\dot{\boldsymbol{\omega}}_{\bar{b}}^{\bar{b}/r}. \tag{42}$$

III. Control Problem

Building on the tracking error dynamics derived in Section II-D, this section formulates the control strategy for the MTAV during the transitional flight phase. Subsection III-A defines the control objectives and presents the proposed control law. Subsection III-B then formulates the control allocation problem distributing efforts among the effectors through constrained quadratic optimization.

A. Control Law Design

Using (35), (36), (40) and (42), we can rearrange those in the usual state-space equations, separating between the attitude and translational tracking errors, which is useful for control law design. Let $\mathbf{x} \triangleq [\mathbf{x}_1\ \mathbf{x}_2\ \mathbf{x}_3\ \mathbf{x}_4]^T$, where $\mathbf{x}_1 \triangleq \tilde{\boldsymbol{\alpha}}$, $\mathbf{x}_2 \triangleq \tilde{\boldsymbol{\omega}}$, $\mathbf{x}_3 \triangleq \tilde{\mathbf{r}}$ and $\mathbf{x}_4 \triangleq \tilde{\mathbf{v}}$, $\mathbf{u}_f \triangleq \bar{\mathbf{f}}_r^c$ and $\mathbf{u}_\tau \triangleq \bar{\boldsymbol{\tau}}_b^{c/B}$, then

$$\dot{\mathbf{x}}_1 = \mathbf{f}_1(\mathbf{x}), \tag{43}$$

$$\dot{\mathbf{x}}_2 = \mathbf{f}_2(\mathbf{x}) + \mathbf{B}_\tau(\mathbf{x})(\mathbf{u}_\tau + \mathbf{d}_\tau), \tag{44}$$

$$\dot{\mathbf{x}}_3 = \mathbf{f}_3(\mathbf{x}), \tag{45}$$

$$\dot{\mathbf{x}}_4 = \mathbf{f}_4(\mathbf{x}) + \mathbf{B}_f(\mathbf{x})(\mathbf{u}_f + \mathbf{d}_f), \tag{46}$$

where $\mathbf{d}_f \triangleq \mathbf{f}_r^{a_{uc}}$ and $\mathbf{d}_\tau \triangleq \check{\boldsymbol{\tau}}$ given by,

$$\check{\boldsymbol{\tau}} \triangleq \check{\boldsymbol{\tau}}_b^d + \left[\left(\mathbf{J}_b^t\left(\tilde{\boldsymbol{\omega}} + \tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r}\right)\right)\times\right]\tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r} + \mathbf{J}_b^t\left[\tilde{\boldsymbol{\omega}}\times\right]\tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r} - \left[\tilde{\boldsymbol{\omega}}\times\right]\mathbf{J}_b^t\tilde{\mathbf{D}}\boldsymbol{\omega}_{\bar{b}}^{\bar{b}/r} - \mathbf{J}_b^t\tilde{\mathbf{D}}\dot{\boldsymbol{\omega}}_{\bar{b}}^{\bar{b}/r}, \tag{47}$$

and $\mathbf{f}_1(\mathbf{x}) \triangleq \mathbf{A}\tilde{\boldsymbol{\omega}}$, $\mathbf{f}_2(\mathbf{x}) \triangleq \left(\mathbf{J}_b^t\right)^{-1}\left[\left(\mathbf{J}_b^t\tilde{\boldsymbol{\omega}}\right)\times\right]\tilde{\boldsymbol{\omega}}$, $\mathbf{f}_3(\mathbf{x}) \triangleq \tilde{\mathbf{v}}$, $\mathbf{f}_4(\mathbf{x}) \triangleq -g\mathbf{e}_3 - \ddot{\bar{\mathbf{r}}}_r^{B/O}$, $\mathbf{B}_\tau(\mathbf{x}) \triangleq \left(\mathbf{J}_b^t\right)^{-1}$ and $\mathbf{B}_f(\mathbf{x}) \triangleq \frac{1}{m}\left(\mathbf{D}^{\bar{b}/r}\right)^T\tilde{\mathbf{D}}^T$.

The control objective is to find $\bar{\mathbf{f}}_r^c = \mathbf{g}_1(\tilde{\mathbf{r}}, \tilde{\mathbf{v}})$ and $\bar{\boldsymbol{\tau}}_b^{c/B} = \mathbf{g}_2(\tilde{\boldsymbol{\alpha}}, \tilde{\boldsymbol{\omega}})$ that stabilizes the aircraft. Therefore, to tackle the problem, a cascaded controller is adopted with an attitude and a position controller, both with a proportional-derivative dynamic inversion given by, using the definitions from (43)–(46),

$$\mathbf{u}_f = -m\left(\mathbf{f}_4(\mathbf{x}) + \boldsymbol{\gamma}_f\right) - \mathbf{f}_r^{a_{uc}}, \tag{48}$$

where $\boldsymbol{\gamma}_f = \mathbf{K}_3\mathbf{x}_3 + \mathbf{K}_4\mathbf{x}_4$, m is the mass of the MTAV, and $\mathbf{K}_3 \in \mathbb{R}$ and $\mathbf{K}_4 \in \mathbb{R}$ are the control law gains. Using the obtained $\mathbf{u}_f$, and reference external angle commands $\bar{\psi}$ and $\bar{\theta}$ we define a new reference frame $\mathcal{S}_d \triangleq \{\vec{x}_d, \vec{y}_d, \vec{z}_d\}$, given by $\mathbf{D}^{d/r} = \mathbf{D}_2(\bar{\theta})\mathbf{D}_3(\bar{\psi})$, that will be used so $\bar{\mathbf{f}}_r^c$ belongs to the $\vec{x}_b$-$\vec{z}_b$ plane. This leads to the new computed required force $\bar{\mathbf{f}}_d^c$, which then gives us a required rotation of $\bar{\phi}$, which will give us the required attitude reference $\mathbf{D}^{\bar{b}/r}$, as follows $\bar{\mathbf{f}}_d^c = \mathbf{D}^{d/r}\bar{\mathbf{f}}_r^c$, $\bar{\phi} = \text{atan}\left(\mathbf{e}_2^T\bar{\mathbf{f}}_d^c \,/\, \mathbf{e}_3^T\bar{\mathbf{f}}_d^c\right)$ and $\mathbf{D}^{\bar{b}/r} = \mathbf{D}_1(-\bar{\phi})\mathbf{D}^{d/r}$.

With the new attitude reference $\mathbf{D}^{\bar{b}/r}$, the attitude errors $\tilde{\boldsymbol{\omega}}$ and $\tilde{\mathbf{D}}$ are computed, and an attitude control loop is applied using

$$\mathbf{u}_\tau = -\left(\frac{\partial\mathbf{f}_1}{\partial\mathbf{x}_2}B_\tau\right)^{-1}\left(\frac{\partial\mathbf{f}_1}{\partial\mathbf{x}_1}\mathbf{f}_1(\mathbf{x}) + \frac{\partial\mathbf{f}_1}{\partial\mathbf{x}_2}\mathbf{f}_2(\mathbf{x}) + \boldsymbol{\gamma}_\tau\right) - \check{\boldsymbol{\tau}}, \tag{49}$$

where $\boldsymbol{\gamma}_\tau = \mathbf{K}_1\mathbf{x}_1 + \mathbf{K}_2\mathbf{f}_1(\mathbf{x})$, and $\mathbf{K}_1 \in \mathbb{R}$ and $\mathbf{K}_2 \in \mathbb{R}$ are the control law gains.

B. Control Allocation Problem

With both control laws set, a required control $\mathbf{u} = [\mathbf{u}_f\ \mathbf{u}_\tau]^T$ is obtained and a control allocation problem is set. Following (20)–(24), the problem is defined as

$$\begin{aligned} \min_{\mathbf{f}} \quad & \frac{1}{2}\mathbf{f}^T\mathbf{H}\mathbf{f} \\ \text{s.t.} \quad & \mathbf{f} \in \mathcal{F} \end{aligned} \tag{50}$$

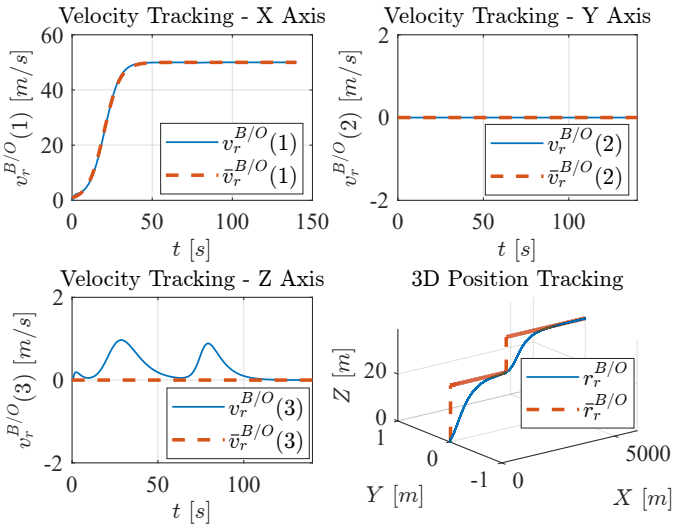


Fig. 4. Achieved and commanded MTAV position and velocity, expressed in the inertial reference frame $\mathcal{S}_r$.

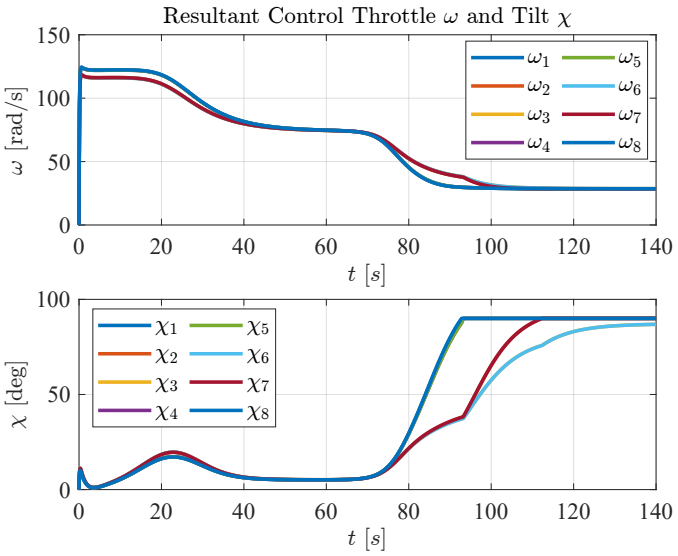


Fig. 5. Achieved propeller angular speed ω $[rad/s]$ and tilt angle χ $[deg]$ for each rotor of the MTAV.

where $\mathcal{F}$ is the viable set where all actuators are respecting their boundaries and achieves the desired control $\mathbf{u}$ and is defined by $\mathcal{F} \triangleq \{\mathbf{f} \in \mathbb{R}^{19} : \boldsymbol{\Theta}\mathbf{f} = \mathbf{u}, \boldsymbol{\Lambda}\mathbf{f} \leq \boldsymbol{\lambda}\}$. This problem only has a solution if $\mathbf{u} \in \mathcal{A}$, where $\mathcal{A}$ is the admissible set and is given by $\mathcal{A} \triangleq \{\mathbf{u} \in \mathbb{R}^6 : \exists \mathbf{f} \in \mathcal{F}^{\text{ineq}}\ s.t.\ \boldsymbol{\Theta}\mathbf{f} = \mathbf{u}\}$, where $\mathcal{F}^{\text{ineq}} \triangleq \{\mathbf{f} \in \mathbb{R}^{19} : \boldsymbol{\Lambda}\mathbf{f} \leq \boldsymbol{\lambda}\}$ is the inequality constraint set, defined by the effectors' boundaries.

IV. Numerical Example

Using the aforementioned modeling, control and control allocation problems, simulations were conducted in MATLAB for a typical MTAV fixed-wing with all propulsion, aerodynamic and other characteristics taken from [5]. The tilting mechanism was assumed to have the same time-response as the aerodynamic surfaces. Boundaries for the effectors were chosen to be $\chi_{max} = \pm 90°$, $\delta_{max} = \pm 30°$, $\omega_{min} = 0$ and $\omega_{max} = 160$ revolutions per second. For the maneuver, a sigmoidal acceleration up to the along heading speed of $u_b = 50\ m/s$ was chosen, and a commanded $\bar{\theta} = -3.65°$ was assumed. For the control allocation cost function, $\mathbf{H} = \mathbf{I}_{19}$ was chosen. As shown in Figs. 4 and 5, the transition between flight regimes is reflected in the evolution of the effector commands. As the aircraft accelerates towards regions where the wing is more effective and, therefore, offers more lift force, the optimization algorithm naturally moves the tilting rotors forward and uses other effectors in order to stabilize the aircraft, correctly performing the transition maneuver.

V. Conclusions

This paper has presented a six-DOF model for the dynamics and kinematics of the aircraft, taking into account the effects of aerodynamic surfaces, rotational inertia of the rotors and tilting mechanisms that allow a full transition of a MTAV from hover to forward-flight. The results obtained show the completeness of the model, especially regarding the tilting movement towards regimes with aerodynamic forces. Many paths can be followed from here to explore different control approaches and control allocation methods, each representing a step toward broader industrial applications for VTOL aircraft and drones.

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References

[1] R. W. Beard and T. W. McLain, Small Unmanned Aircraft: Theory and Practice. Princeton, NJ, USA: Princeton Univ. Press, 2012, pp. 39-59.

[2] J. A. Bezerra and D. A. Santos, "Optimal exact control allocation for under-actuated multirotor aerial vehicles," IEEE Control Syst. Lett., vol. 6, pp. 1448-1453, 2022, doi: 10.1109/LCSYS.2021.3110490.

[3] D. A. Santos, J. F. S. Trentin, J. A. Ricardo Junior, L. G. P. Roéfero, T. R. Oliveira, "Adaptive sliding mode attitude control for multirotor aerial vehicles with experimental evaluation," J. Braz. Soc. Mech. Sci. Eng., vol. 46, art. no. 194, 2024, doi: 10.1007/s40430-024-04786-1.

[4] J. Willis and R. Beard, "Pitch and thrust allocation for full-flight-regime control of winged eVTOL UAVs," IEEE Control Syst. Lett., vol. 6, pp. 1058-1063, 2021, doi: 10.1109/LCSYS.2021.3089130.

[5] G. H. C. Silva, "Design of control laws for the transition phase of a lift-plus-cruise eVTOL aircraft," M.S. thesis, Instituto Tecnológico de Aeronáutica, São José dos Campos, Brazil, 2024, in press.

[6] G. H. C. Silva, J. A. Ricardo Junior, D. A. Santos, and Y. M. Abdallah, "Control and command allocation architectures for takeoff, transition, and horizontal flight of NASA's lift-plus-cruise eVTOL concept," in Proc. 28th ABCM Int. Congr. Mech. Eng. (COBEM), Curitiba, Brazil, 2025, Paper COB-2025-2081.

[7] F. L. Markley and J. L. Crassidis, Fundamentals of Spacecraft Attitude Determination and Control. New York, NY, USA: Springer, 2014, doi: 10.1007/978-1-4939-0802-8.